03 - Literature Review and Research Gap

Only references present in report/references.bib

Knowledge-pack rule: every statement below is derived from files in this project or is explicitly labelled as planned, missing or not verified. Nothing here is an invented citation, dataset, result or feature.

Source files used for this document

report/references.bib; report/chapters/chapter_2_literature_review.md; review/review_1.pptx

These are the works cited in the project's own references.bib. This pack adds no other citations. The bib file itself says to verify each entry against the publisher record before submission; that verification has not been done by this pack.

Themes and how each informs the project

Theme	Cited works (bib keys)	How the project uses it
ML fraud detection	ngai2011application, bhattacharyya2011data, chen2016xgboost, breiman2001random	Logistic Regression, Random Forest, XGBoost as supervised models
Feature engineering	bahnsen2016feature	History-based aggregates vs the customer's prior mean
Imbalance and evaluation	davis2006relationship, saito2015precision, dalpozzolo2015calibrating	PR-AUC emphasis; class weights instead of resampling
Anomaly detection	liu2008isolation, breunig2000lof, chandola2009anomaly	Isolation Forest and LOF
Hyper-parameter search	bergstra2012random	RandomizedSearchCV
Trust in e-commerce	mcknight2002developing, gefen2003trust, pavlou2003consumer, davis1989perceived	Multi-dimensional trust; five survey dimensions
Measurement and privacy	cronbach1951coefficient, likert1932technique, sweeney2002kanonymity	Cronbach's alpha; Likert items; k-anonymity
Tooling	pedregosa2011scikit	scikit-learn

Bibliography as recorded in references.bib

Key	Authors	Title	Year
liu2008isolation	Liu, Fei Tony and Ting, Kai Ming and Zhou, Zhi-Hua	Isolation Forest	2008}, pages = {413--422
breunig2000lof	Breunig, Markus M. and Kriegel, Hans-Peter and Ng, Raymond T. and Sand	LOF: Identifying Density-Based Local Outliers	2000}, pages = {93--104
chen2016xgboost	Chen, Tianqi and Guestrin, Carlos	XGBoost: A Scalable Tree Boosting System	2016}, pages = {785--794

Key	Authors	Title	Year
breiman2001random	Breiman, Leo}, title = {Random Forests	Random Forests	2001}, pages = {5--32
pedregosa2011scikit	Pedregosa, Fabian and others}, title = {Scikit-learn: Machine Learning	Scikit-learn: Machine Learning in Python	2011}, pages = {2825--2830
bergstra2012random	Bergstra, James and Bengio, Yoshua}, title = {Random Search for Hyper-	Random Search for Hyper-Parameter Optimization	2012}, pages = {281--305
saito2015precision	Saito, Takaya and Rehmsmeier, Marc	The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Cla	2015}, pages = {e0118432
davis2006relationship	Davis, Jesse and Goadrich, Mark}, title = {The Relationship Between Pr	The Relationship Between Precision-Recall and ROC Curves	2006}, pages = {233--240
dalpozzolo2015calibrating	Dal Pozzolo, Andrea and Caelen, Olivier and Johnson, Reid A. and Bonte	Calibrating Probability with Undersampling for Unbalanced Classification	2015
bahnsen2016feature	Bahnsen, Alejandro Correa and Aouada, Djamila and Stojanovic, Aleksand	Feature Engineering Strategies for Credit Card Fraud Detection	2016}, pages = {134--142
bhattacharyya2011data	Bhattacharyya, Siddhartha and Jha, Sanjeev and Tharakunnel, Kurian and	Data Mining for Credit Card Fraud: A Comparative Study	2011}, pages = {602--613
ngai2011application	Ngai, E. W. T. and Hu, Yong and Wong, Y. H. and Chen, Yijun and Sun, X	The Application of Data Mining Techniques in Financial Fraud Detection	2011}, pages = {559--569
chandola2009anomaly	Chandola, Varun and Banerjee, Arindam and Kumar, Vipin}, title = {Anom	Anomaly Detection: A Survey	2009}, pages = {1--58
mcknight2002developing	McKnight, D. Harrison and Choudhury, Vivek and Kacmar, Charles	Developing and Validating Trust Measures for e-Commerce	2002}, pages = {334--359
gefen2003trust	Gefen, David and Karahanna, Elena and Straub, Detmar W.	Trust and TAM in Online Shopping: An Integrated Model	2003}, pages = {51--90
pavlou2003consumer	Pavlou, Paul A.}, title = {Consumer Acceptance of Electronic Commerce	Consumer Acceptance of Electronic Commerce	2003}, pages = {101--134
davis1989perceived	Davis, Fred D.}, title = {Perceived Usefulness, Perceived Ease of Use,	Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology	1989}, pages = {319--340

Key	Authors	Title	Year
cronbach1951coefficient	Cronbach, Lee J.}, title = {Coefficient Alpha and the Internal Structu	Coefficient Alpha and the Internal Structure of Tests	1951}, pages = {297--334
likert1932technique	Likert, Rensis}, title = {A Technique for the Measurement of Attitudes	A Technique for the Measurement of Attitudes	1932}, pages = {1--55
sweeney2002kanonymity	Sweeney, Latanya}, title = {k-Anonymity: A Model for Protecting Privac	k-Anonymity: A Model for Protecting Privacy	2002}, pages = {557--570

Research gap (as stated in the project)

- Strong fraud models are usually evaluated in isolation.
- Trust surveys are run without security telemetry.
- Monitoring is treated as an operations topic, not as a research instrument.
- Missing: a joint view of measured security and perceived trust, a transparent and sensitivity-tested composite index, and a reproducible, observable research platform.

Position of the project against the gap

Gap element	Project response	Status
Joint view of security and trust	Platform Security Index + Consumer Trust Score -> Consumer Trust Index	IMPLEMENTED
Sensitivity-tested index	notebook 05: survey weight sweep and random dimension weights	IMPLEMENTED
Validated index	No psychometric validation (e.g. factor analysis) done	MISSING
Reproducible, observable platform	Docker Compose + Prometheus/Grafana/Loki	IMPLEMENTED
Review of literature beyond the bib list	chapter_2_literature_review.md is a short draft (~7 sections)	PARTIAL

Missing documentation: the literature review draft does not yet compare studies in depth (datasets, reported metrics) and contains no table of reviewed papers' results. Do not invent such comparisons.